MENDELIAN INHERITANCE:

A. Terminology:

- Gene – Specific sequence of nucleotides on the chromosome (DNA) that governs ~~a particular~~ or encodes particular proteins, which in turn express in the form of some particular feature of the organism.
- Allele – The alternative form of a gene, occupying the same position on a chromosome and effecting the same characteristic but in different ways.
- Dominant factor/allele – It is the allele which express itself in a hybrid, where the members of a pair of alleles for the character are different
- Recessive factor / allele – It is the allele which ~~express itself in a~~ is suppressed in the hybrid, where the members of a pair of alleles for the character are different
- Phenotype – The observable or external characteristics of an organism which is genetically controlled.
- Genotype – The genetic constitution of an organism
- Homozygous / Pure – The condition in which a pair of chromosomes carries similar alleles of a particular character
- Heterozygous / Hybrid – The condition in which a pair of chromosomes carries dissimilar alleles for a particular character.
- Monohybrid cross – Cross made between two individuals of a species, considering the inheritance of the contrasting pair of a single character.
- Dihybrid cross – Cross made between two individuals of a species, considering the inheritance of the contrasting pair of two character.
- Filial generation – Generation next to parent.
- Back cross – Cross between F_1 individual and any of the parent.
- Test cross – Cross between F_1 individual and the recessive parent.

Q. Why Mendel is called 'Father of Genetics'?

A: Gregor Johann Mendel (1822-84) was the first to explain the mechanism of transmission of characteristics from parent to offspring generation after generation. His findings were published in 'Proceedings of Natural History', 1865.

Q. Who rediscovered Mendel's work?

A: Mendel's findings were not recognised then since his methodology during the investigation of the inheritance pattern included mathematical logic and statistical analysis of data.

Later, 16 years after his death, his findings were rediscovered and brought under limelight by Hugo de Vries Correns and Tschermark in the year 1900.

Q. Why did Mendel choose garden pea (Pisum sativum) for his experiments?

A: Following are the reasons for choosing garden pea -

- It is an annual plant, thus can be grown fast.
- Flower, complete and self-pollinating, thus varieties were available in pure forms, i.e, produced the same type generation after generation.
- It has a number of easily detectable contrasting character.
- Though normally self-pollinated yet can also be cross-pollinated artificially.

D. Mendel's laws and their mathematical elucidation:

- **1st Law - Law of Dominance**

Out of a pair of contrasting characters present together, only one is able to express itself while the other remains suppressed. The one that expresses is the dominant trait and the one which is suppressed is the recessive trait.

Mathematical elucidation:

Character : Plant height of garden pea
Gene : Gene for plant height
Allele : Dominant - Tall (T)
Recessive - Dwarf (t)

Parent : ♂ (TT) × ♀ (tt)

Gamete : (T) (t)

F1 gen : Tt.

Phenotype - 100% Tall
Genotype - 100% Heterozygous tall.

} 1st law proved

- 2nd law - Law of segregation / Law of purity of gametes.

The members of the allelic pair that remain together in the parent, do not mix-up or blend, but segregate/separate during gamete formation and only one of the factors enter a gamete.

Mathematical elucidation [Monohybrid Cross]

Character : Plant height of garden pea
Gene : Gene for plant height
Allele : Dominant - Tall (T)
Recessive - Dwarf (t).

Parent : ♂ (TT) × ♀ (tt)

Gamete : (T) (t)

F1 gen. : Tt.

Phenotype - 100% Tall
Genotype - 100% Heterozygous tall.

Selfing : F1 × F1

♂ (Tt) ♀ (Tt)

Gamete : (T) (t) (T) (t)

} 2nd law proved

Punnett square :

♀ \ ♂	T	t
T	TT	Tt
t	Tt	tt

F_2 gen :

Phenotype : Tall - 3/4
Dwarf - 1/4.

Phenotypic ratio = Tall : Dwarf :: 3 : 1.

Genotype : Homozygous tall (TT) - 1/4
Heterozygous tall (Tt) - 2/4
Homozygous dwarf (tt) - 1/4.

Genotypic ratio = TT : Tt : tt :: 1 : 2 : 1

- 3rd law - Law of independent assortment

When there are two pairs of contrasting characters, the distribution of the members of one pair into the gamete is independent of the distribution of the other pair.

Mathematical elucidation [Dihybrid cross].

Character 1 : Plant height of garden pea
Gene : Gene for plant height.
Allele : Dominant - Tall (T)
Recessive - Dwarf (t)
Character 2 : Seed shape of garden pea
Gene : Gene for seed shape
Allele : Dominant - Round (R)
Recessive - Wrinkled (r).

Parent : ♂ (TTRR) × ♀ (ttrr).

Gamete : (TR) (tr)

F_1 gen : TtRr.

Phenotype - 100% Tall and Round
Genotype - 100% heterozygous tall and round

Selfing : F_1 × F_1

♂ (TtRr) ♀ (TtRr)

Gamete : (TR) (Tr) (tR) (tr) (TR) (Tr) (tR) (tr)

} 3rd law proved

Punnett square:

♀ \ ♂	TR	Tr	tR	tr
TR	TTRR	TTRr	TtRR	TtRr
Tr	TTRr	TTrr	TtRr	Ttrr
tR	TtRR	TtRr	ttRR	ttRr
tr	TtRr	Ttrr	ttRr	ttrr

F2 gen. :

Phenotype : Tall and Round - 9/16

Tall and Wrinkled - 3/16

Dwarf and Round - 3/16

Dwarf and Wrinkled - 1/16.

Phenotypic ratio = 9 : 3 : 3 : 1.

Q. What is a Test cross ? Discuss its significance.

A: It is a special type of back cross which is made between individual with a dominant trait and its recessive parent in order to know whether the individual is homozygous or heterozygous for the trait. Since the cross is used to determine the genotype of an individual, it is called a TEST CROSS.

Offspring of a test cross will be 100% dominant if the individual is homozygous dominant. On the other hand, the progeny will consist of 50% dominant and 50% recessive trait if the individual is heterozygous dominant.

If the individual (under discussion) is heterozygous dominant then in case of monohybrid, the test cross will give a ratio of 1 : 1 and in case of dihybrid, the test cross will give a ratio of 1 : 1 : 1 : 1.

Monohybrid test cross

Test cross I:

Individual - Heterozygous tall

Individual (Tt) × Recessive parent (tt)

Gamete: (T) (t) (t)

Progeny: Tall - 50%
Dwarf - 50%

Test cross ratio = 1 : 1.

Test cross II:

Individual - Homozygous tall

Individual (TT) × Recessive parent (tt)

Gamete: (T) (t)

Progeny: Tall - 100%

Dihybrid test cross

Test cross I

Individual - Heterozygous tall and round

Individual (TtRr) × Recessive parent (ttrr)

Gamete: (TR) (Tr) (tR) (tr) (tr)

Progeny: Tall and Round - 25%. Dwarf and Round - 25%.
Tall and Wrinkled - 25%. Dwarf and Wrinkled - 25%.

Test cross ratio = 1 : 1 : 1 : 1.

Test cross II

Individual - Homozygous tall and round

Individual (TTRR) × Recessive parent (ttrr)

Gamete: (TR) (tr)

Progeny: Tall and Round - 100%.

DEVIATIONS FROM MENDELISM:

A. Incomplete dominance

Sometimes in a heterozygote, dominant alleles does not completely mask the phenotypic expression of the recessive allele and there occurs an intermediate phenotype in the heterozygote. This is called incomplete dominance

Eg:—

Character : Flower colour in Snapdragon
Gene : Gene for flower colour
Allele : Dominant - Red (R)
Recessive - White (r)

Parent : ♂ (RR) × ♀ (rr)

Gamete : (R) (r)

F_1 gen : Rr

Phenotype - 100% Pink
Genotype - 100% Heterozygous pink

Selfing : F_1 × F_1

Rr × Rr.

Gamete : (R) (r) (R) (r)

Punnett square :

♀ \ ♂	R	r
R	RR	Rr
r	Rr	rr

F_2 gen :

Phenotype : Red flower (1/4); Pink (2/4); White (1/4)
Phenotypic ratio = Red : Pink : White :: 1 : 2 : 1
Genotype : Homozygous Red (1/4); Heterozygous pink (2/4); Homozygous white (1/4).

Genotypic ratio = RR : Rr : rr :: 1 : 2 : 1.

B. Co-dominance:

Sometimes both the alleles of a gene in a heterozygote lack the dominant recessive relationship, i.e, each allele is capable of some degree of phenotypic expression. This is called co-dominance.

Eg:—

Character : Coat colour of shorthorn breed cattle.
Gene : Gene for coat colour.
Allele : Red coat colour (C^R)
White coat colour (C^W).

Parent : ♂ (C^RC^R) × ♀ (C^WC^W)

Gamete : (C^R) (C^W)

F_1 gen : C^RC^W

Phenotype : 100% Roan coat

Genotype : 100% Heterozygous roan coat.

Sib-mating : F_1 × F_1

(C^RC^W) (C^RC^W)

Gamete : (C^R) (C^W) (C^R) (C^W)

Punnett square:

♀ \ ♂	C^R	C^W
C^R	C^RC^R	C^RC^W
C^W	C^RC^W	C^WC^W

F_2 gen :

Phenotype : Red coat (1/4); Roan coat (2/4); White coat (1/4)

Phenotypic ratio = Red coat : Roan coat : White coat :: 1 : 2 : 1.

Genotype : Homozygous red coat (1/4); Heterozygous roan coat (2/4); Homozygous white coat (1/4).

Genotypic ratio = $C^RC^R : C^RC^W : C^WC^W$ :: 1 : 2 : 1.

Chromosome Mapping [Gene mapping].

The distance between two points on the genetic map of a chromosome gives the average number of crossovers between them. One map unit (m.u.) is equal to one percent of recombinant phenotypes. A map unit is called a centiMorgan (cM) in T. H. Morgan's honor. 100 centiMorgans equal one Morgan (M).

Importance of Chromosome Maps

1. The chromosome maps give information on average number of crossovers for a population, recombinant frequency, number of chiasmata, etc.
2. Genetic map distances can be determined from the frequency of recombination between genes in experimental crosses.

3. Comparing map distance obtained from frequency of recombination and the actual map distance, the prediction of multiple cross over events can be made.
4. The actual gene order can be found out from the phenotype classes of the F_2 generation in experimental crosses employed for determining map distance.

Non-allelic or Inter-genic Gene Interaction

1. **Complementary genes (9:7 ratio):** The complementary genes are two pairs of non-allelic dominant genes (i.e., present on separate gene loci), which interact to produce only one phenotypic trait, but neither of them if present alone produces the phenotypic trait in the absence of the other. For example, gene for flower color in sweet pea. The flowers in this plant are either purple (if they contain anthocyanin pigment) or white. Two independently assorting genes (*C* and *P*) are involved in anthocyanin synthesis and each gene has a recessive allele to inhibit pigment production. When either of the two genes or both are present, purple flowered plants were obtained; however, the presence of one alone is not sufficient to produce colored varieties. On self-crossing, the hybrid, they produce a ratio of 9 (Purple):7 (White) in F_2 generation (Figure 3).

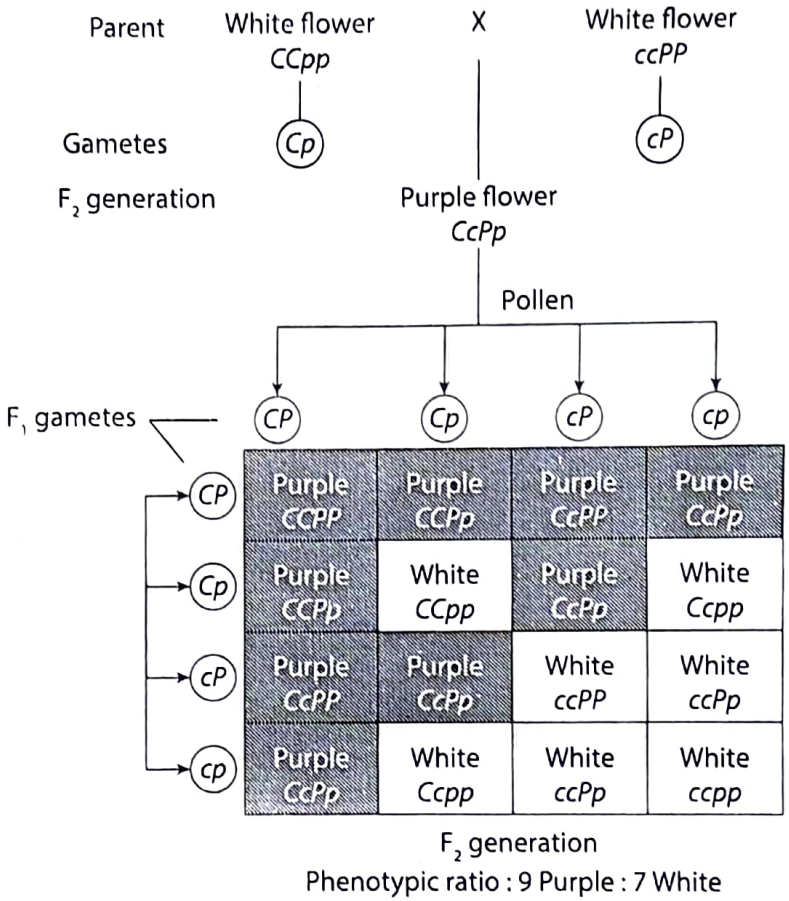


Figure 3 Crosses of complementary genes in sweet pea plant.

2. **Supplementary genes (9:3:4 ratio):** Supplementary genes are two independent pairs of dominant genes which interact in such a way that one dominant gene will produce its effect whether the other is present or not, for example, combs in chicken. When Wyandottes (rose combs) were crossed with Brahmas (pea comb, another breed of chickens were produced which had a "walnut" shaped comb. Wyandottes have the

genotype *RRpp*, while Brahmas have the genotype *rrPP*. The F_1 hybrids between these two varieties have walnut combs with genotype *RrPp*. When these hybrids were intercrossed with each other, all four types of combs appeared in the progeny in the ratio of: 9 (Walnut): 3 (Rose): 3 (Pea): 1 (Single) (Figure 4).

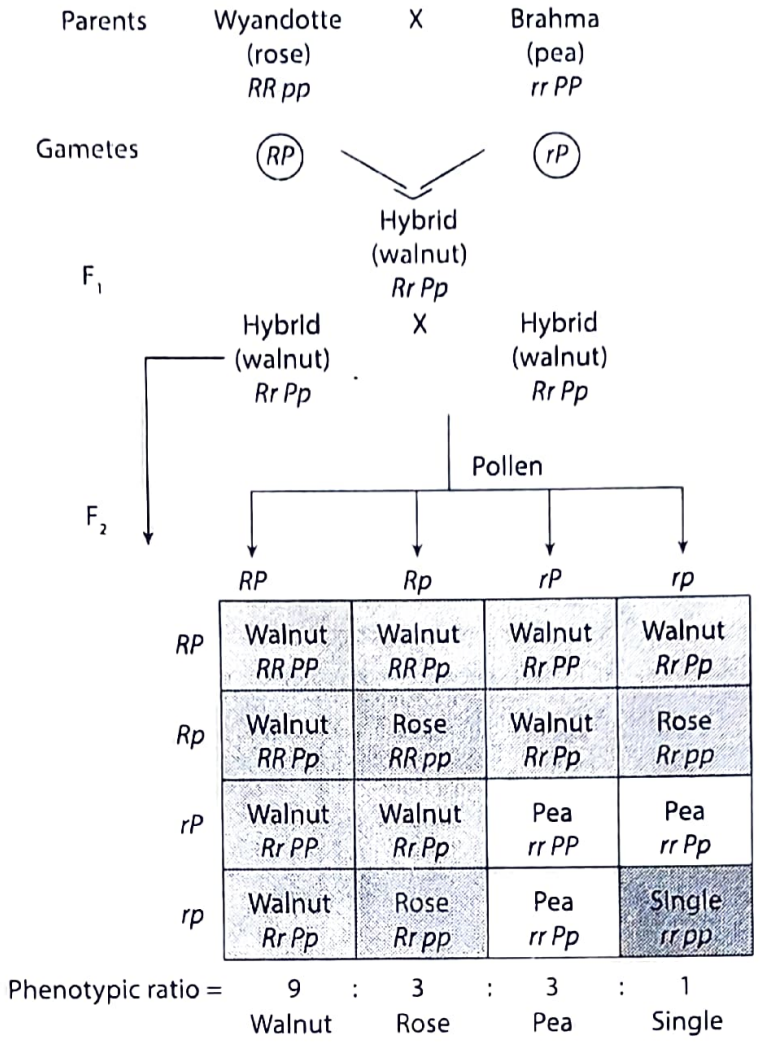


Figure 4 Crosses of supplementary genes in chickens.

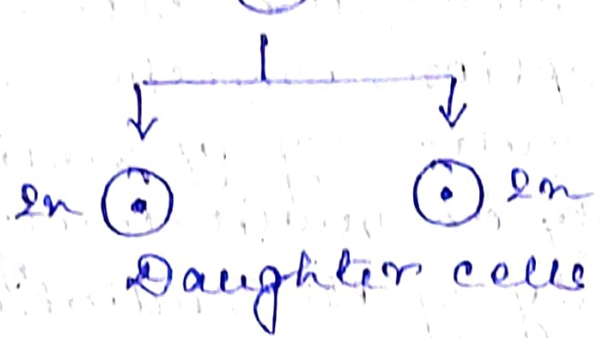
3. **Epistasis (Inhibiting genes):** Epistasis is the interaction between non-allelic genes (present on separate loci) in which one-gene masks, inhibits or suppresses the expression of other gene. The gene that suppresses the other gene is known as inhibiting or epistatic factor and the one, which is prevented from exhibiting itself, is known as hypostatic. Although, it is similar to dominance and recessiveness, but the two factors occupy two different loci. Epistasis can be of the following types:
   (a) Dominant epistasis (12:3:1 or 13:3 ratio): In dominant epistasis out of two pairs of genes the dominant allele, (i.e., gene A) of one gene masks the activity of other allelic pair (Bb). Dominant epistasis in dogs.
   (b) Recessive epistasis (9:3:4 ratio): Epistasis due to recessive gene is known as recessive epistasis, that is, out of the two pairs of genes, the recessive epistatic gene masks the activity of the dominant gene of the other gene locus, for example, in mice agouti colour.
4. **Duplicate genes (15:1 ratio):** Sometimes two pairs of genes located on different chromosomes determine the same phenotype. These genes are said to be duplicate of each other. For example, in shepherd's purse (*Bursa bursapastoris*), the seed capsules have two shapes—triangular and oval. If the genotype is *aabb*, that is, only recessive alleles in both genes are present, then oval seed capsules were formed. However, if the dominant allele of either gene is present, triangular seed capsules are formed.

Topic : Mitosis and Meiosis to be taught as a part of genetics.

A. Mitosis :

The process of cell division in which two daughter nuclei are formed identical to mother nucleus with equal number of chromosome and equal quantity of cytoplasm. It is also known as equational division or somatic cell division.

~~Occurrence~~



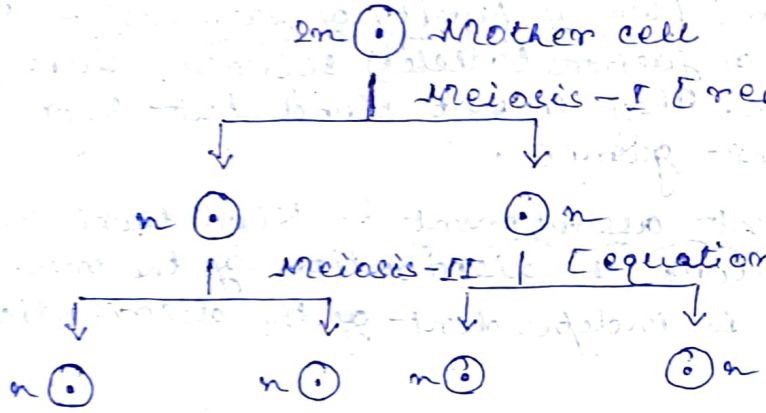
Occurence : • In the growing region of apical part of stem, root and growing leaves in plants.

- In the time of foetus development, growth and repair of the body in higher animals.
- During vegetative and asexual reproduction

Significance : • The number of cells within an organism increases by mitosis.
• Cell replacement and repair.
• Regeneration of lost parts.
• Some lower group of organisms produce genetically similar progeny through asexual reproduction by mitosis.

B. Meiosis :

It is a special type of cell division by which a reproductive cell gives rise to four haploid daughter cells, each having exactly half of the number of chromosomes present in the mother cell.



Occurence : • In lower plants meiosis takes place in spore mother cells (2n) giving rise to haploid (n) spores.
• In higher plants, it takes place in pollen mother cells of antheridium and egg mother cells of ovule for the formation of pollen and ovum respectively
• In higher animals, meiosis takes place in primary gametocytes during gamete formation

Significance : • Meiosis maintains the fixed number of chromosomes in sexually reproducing organisms. It is essential since the chromosome number becomes double after fertilization
• Meiosis generates genetic variation in the diploid cell, in part by the exchange of genetic information between the pairs of chromosomes after they recombine
• Meiosis is essential in sexually reproducing organisms as it forms gametes.

Mendelian Disorders [single gene disorders in human].

These disorders are inherited to the offspring following principle of inheritance and are mainly caused by alterations or mutation in alleles of a single gene.

1. Hemophilia is an X-linked disorder in which persons are unable to produce a factor for blood clotting. The principal type of hemophilia in humans is due to a recessive X-linked mutation, and it is most common in males.
2. Color blindness or color vision deficiency is decreased ability to perceive color differences; under normal light conditions. Color blindness is usually a sex-linked condition.
3. Sickle cell anemia is an autosomal recessive disorder that is inherited. People with two sickle cell genes have severe anemia; those with only one defective gene have the sickle cell trait.
4. Phenylketonuria or PKU is a genetic error of protein metabolism characterized by elevated blood levels of the amino acid phenylalanine.
5. Thalassemia is an autosomal recessive form of anemia which occurs due to mutation or deletion of the genes controlling the synthesis of globin chains of hemoglobin. It is mainly of three types—α-Thalassemia, β-Thalassemia, and δ-Thalassemia.
6. Alkaptonuria is an autosomal recessive genetic disease caused by problems in phenylalanine and tyrosine metabolism. It is caused due to a defect in the enzyme homogentisate 1,2-dioxygenase, which catalyzes the degradation of tyrosine.
7. Polydactyly is condition in which the individual develops extra fingers or toes. It is caused by incomplete penetrance of trait, that is, when individuals do not show a trait even though they have the appropriate genotype.

8. Gaucher's disease is an autosomal recessive disorder is caused by deficiency of enzymes lysosomal exoglycosidase and β-glucocerebrocidase which leads to accumulation of glucocerebrocides in white blood cells, causing malfunction of spleen, liver, kidneys, lungs, brain and bone marrow.
9. Tay Sach's disease is caused by a mutant allele located on HEXA gene, which also encodes the enzyme hexosaminidase A and is inherited to homozygous individuals. It leads to accumulation of ganglioside GM2 and blockage of the action of neurons, leading to deterioration of the nervous system and eventually paralysis.
10. Huntington's disease is a neurodegenerative disorder caused due to dominant gene (HD gene) on chromosome 4.
11. Cystic fibrosis is an autosomal recessive mutation of CF gene and leads to abnormal mucus in the lungs, pancreas, and sweat glands.